

Mohammed Zuhaib Damudi

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Education

Northeastern University

Master of Science in Computer Science

Sep 2025 – May 2027

Boston, MA

- **Coursework:** Algorithms, Database Management Systems, Foundations of AI, Programming Design Paradigm

Manipal Institute of Technology

Bachelor of Technology in Computer Science and Engineering, Minor Specialization in Big Data Analytics

Jul 2019 - Jul 2023

Manipal, India

- **Research Thesis:** "Single-Step Sampling Approach for Unsupervised Anomaly Detection of Brain MRI Using Denoising Diffusion Models" - *International Journal of Biomedical Imaging*, Vol 2024, Issue 1; DOI: [10.1155/ijbi/2352602](https://doi.org/10.1155/ijbi/2352602)

Experience

Shell

Software QA Engineer

Aug 2023 – Aug 2025

Bengaluru, India

- Ran performance tests across **10+ projects** in Finance, Workforce Management, and IAM, simulating up to **200 req/s** and **3,000 concurrent users** to find bottlenecks before production.
- Replicated real-world production workloads with live data loads across SAP systems (CFIN, BW, PaPM), exposing memory and CPU bottlenecks that caused system crashes under load.
- Identified redundant API calls on a web application where endpoints were being triggered multiple times per action, cutting unnecessary network requests by over **50%**.
- Contributed to a patent-filed *AI-powered code converter tool (GPT-4o)* that automated **~80%** of manual programming effort.

Questt AI

Data Engineer Intern

May 2022 – Jul 2022

Bengaluru, India

- Designed and implemented an ETL pipeline in Apache Airflow that extracted and flattened over **1 million** heavily nested MongoDB documents into a structured relational schema in Google BigQuery.
- Implemented dynamic detection and adaptation for upstream MongoDB schema changes, eliminating the need for manual pipeline updates.
- Built the company's primary data warehouse in BigQuery, replacing a third-party vendor and giving analysts direct SQL access for ad-hoc querying and reporting.

Projects

Publication - Anomaly Detection in Brain MRI using Generative AI | Python, PyTorch, Matplotlib

Jan 2023 - Jul 2023

- Proposed a novel single-step sampling approach for brain tumor detection using DDPMs, reducing inference time from **~50s** to **~0.24s** per image. Published as first author in Wiley Online Library.
- Introduced a partial diffusion technique to generate healthy brain MRI reconstructions, detecting anomalous regions with approximately **80%** accuracy.
- Achieved up to **66% improvement** in segmentation metrics (IoU, Precision, Recall, AUC) over the baseline iterative sampling approach.

Runalytics - Strava Running Analytics Dashboard | Python, NumPy, Pandas, Scikit-learn, Flask

Dec 2025 - Jan 2026

- Processed Strava running exports containing heart rate, elevation, cadence, and GPS data, extracting **30+ performance metrics** and generating pace improvement trends and PR tracking.
- Categorized runs into their respective training types using rule-based classification on pace and distance patterns.
- Developed an interactive dashboard with synchronized chart-to-map hover tracking, city-based route filtering via reverse geocoding, and dynamic elevation and heart rate visualizations.

Habitual - Mobile Habit Tracker Application | Flutter, SQLite

Jan 2022 - Mar 2022

- Built and published a cross-platform habit-tracking app in Flutter on the Google Play Store, reaching **100+ downloads**.
- Tracked streaks, weekly progress, and completion rates for each habit, with SQLite for local data storage.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, HTML/CSS, Dart, SQL

Technologies: NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn, OpenCV, MongoDB, MySQL, Kibana, Git, Docker, Flutter, Google BigQuery, Apache Airflow, Apache JMeter, LoadRunner Enterprise, Azure Load Testing, Selenium, Postman

Concepts: Data Science, Data Engineering, Machine Learning, Big Data Analytics, Data Visualization, Anomaly Detection, Computer Vision, Generative AI, ETL Pipelines, Automation